

How to Choose Your Kubernetes Deployment Model

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By Analysts [Wataru Katsurashima](#), [Arun Chandrasekaran](#), [Dennis Smith](#)

There are various deployment models of Kubernetes, from self-managed software solutions to provider-managed cloud services, each being a correct choice for different scenarios. I&O leaders should use this research note to understand trade-offs of each model and to determine a right deployment model.

Overview

Key Challenges

- Evaluation criteria for container platforms aren't commonly found within enterprise IT, due to the lack of experience of container management.
- There are various deployment models of Kubernetes, from do-it-yourself (DIY) open source to commercially supported software solutions and cloud services, each with significant implications on costs, risks and skills required.
- I&O leaders are required to quickly respond to a development team's needs for Kubernetes, despite the rapidly evolving market and the lack of maturity of container ecosystem.
- Management capabilities needed for production systems, such as security, monitoring and storage/networking integration are not "built in" to Kubernetes. I&O has to have a plan to fill the gap.

Recommendations

I&O leaders responsible for infrastructure agility should:

- Collaborate with application leaders to identify the organization's priorities in deploying Kubernetes, based on seven criteria.
- Understand the trade-offs of each deployment model for the organization by using Gartner's assessment result adjusted according to the organization's priorities and contexts.
- Leverage the cloud's advantages of operational simplicity, speed and cost by considering using public cloud container services as an initial choice.
- Consider using container management software to better support hybrid cloud and multicloud use cases at production phase. Managed services by third-party providers or hosted services

are good alternatives to container management software to alleviate the operational burden.

Introduction

As Kubernetes becomes the de facto standard in container orchestration, application development teams at enterprises are beginning to demand production Kubernetes environments. However, there are various deployment models of Kubernetes, from do-it-yourself open source to commercially supported software solutions and cloud services (see “Market Guide for Container Management Software” and “Market Guide for Public Cloud Container Services”), each with significant implications on costs, risks and skills required. If I&O does not have appropriate priorities or criteria, there is a risk of making decisions that are not aligned with the company's IT strategy. (See Note 1 for sample vendors that are active in the Kubernetes ecosystem.)

This research clarifies trade-offs of the various options, based on the definition of the deployment models shown in Table 1. It then provides best practices for I&O leaders to determine, with application leaders, a right deployment model.

Figure 1. Kubernetes Deployment Models

Kubernetes Deployment Models				
Deployment Model	1. Public Cloud Container Services	2. Container Management Software		3. Do-It-Yourself With Upstream Version
Description	A deployment model where users consume a managed Kubernetes service offered by a public cloud IaaS provider. Users don't have to build or manage Kubernetes clusters but cannot flexibly select configurations such as Kubernetes version.	A deployment model where users build and manage Kubernetes clusters on-premises and/or off-premises by using a packaged software solution. The software solutions combine a Kubernetes distribution with other management capabilities, such as security, monitoring and storage/networking integration. Gartner identified two major types of product approaches in this model (see below).		A deployment model where users build and manage Kubernetes clusters on-premises and/or off-premises by using the open-source Kubernetes upstream version from its GitHub repository. This has been a common deployment model among early adopters.
		2a. Container Operations Focus This approach focuses on simplifying container operations management, providing more vendor neutrality in DevOps toolchains and application infrastructure software.	2b. Kubernetes-Based Application Platform This approach provides a more comprehensive DevOps and microservice development experience by including DevOps toolchains and application infrastructure software for adjacent middleware services.	
Examples	- Amazon Elastic Container Service for Kubernetes (Amazon EKS) - Azure Kubernetes Service (AKS) - Google Kubernetes Engine (GKE)	- Docker Enterprise Edition - Heptio - Mesosphere Enterprise DC/OS - Rancher	- Pivotal Cloud Foundry - Red Hat OpenShift Container Platform - SUSE Cloud Application Platform	Kubernetes hosted by the Cloud Native Computing Foundation (CNCF)

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Source: Gartner (August 2018)

Note: there are two other deployment models not included in Figure 1: managed services by third-party providers, and hosted services. Analysis on these are included in the Understand the Trade-Offs of Each Deployment Model section.

Analysis

Identify Priorities Based on These Seven Criteria

Evaluation criteria for container platforms aren't commonly found within enterprise IT due to the lack of container management experience.

As the first step in evaluating Kubernetes deployment options, Gartner recommends that I&O leaders collaborate with application leaders to identify priorities of their organization's Kubernetes deployment project, along with the key decision criteria outlined below.

Table 1: Seven Criteria for Kubernetes Deployment

Criteria ↓	Description ↓	Why Important ↓
Operational Simplicity	How much the steep learning curve of Kubernetes can be moderated.	Kubernetes operations require a very different set of technologies and knowledge from existing systems. It is a key challenge for many I&O organizations to overcome the steep learning curve.
Speed	How quickly Kubernetes clusters can be deployed and updated.	I&O leaders need to timely meet the application development team's request.
Comprehensive Management	How easily comprehensive management capabilities for production systems can be built.	Integrating management capabilities not built into Kubernetes, such as security, monitoring and storage/networking integration, is necessary for production systems. It is commonly a significant challenge for I&O.
Hybrid Cloud and Multicloud Support	How well hybrid cloud or multicloud use cases can be supported.	Application portability across hybrid cloud or multicloud is often a major requirement in using containers.
DevOps and Microservices Support	How well DevOps toolchains and microservices application infrastructure are integrated.	DevOps and microservices application development are the two most common use cases for Kubernetes.

Criteria ↓	Description ↓	Why Important ↓
Flexibility and Customizability	How flexibly operation tools can be chosen; how much customization of Kubernetes is possible.	Since new operations tools are released frequently in this area, it is important to be able to flexibly choose tools. Kubernetes is extensible through plugins or other mechanisms, but there are still use cases where customization is necessary for better performance or integration with existing systems.
Cost Structure	What the cost structure of each option is.	A major cost factor is very different among deployment models (e.g., human resource, product); I&O leaders need to understand the difference.

Source: Gartner (August 2018)

Priorities should vary according to the organization's situation. For example, some organizations might prioritize operational simplicity for talent shortage. Some might put an emphasis on hybrid cloud because a cloud migration project is ongoing. I&O leaders need to understand what factors are important to their organizations.

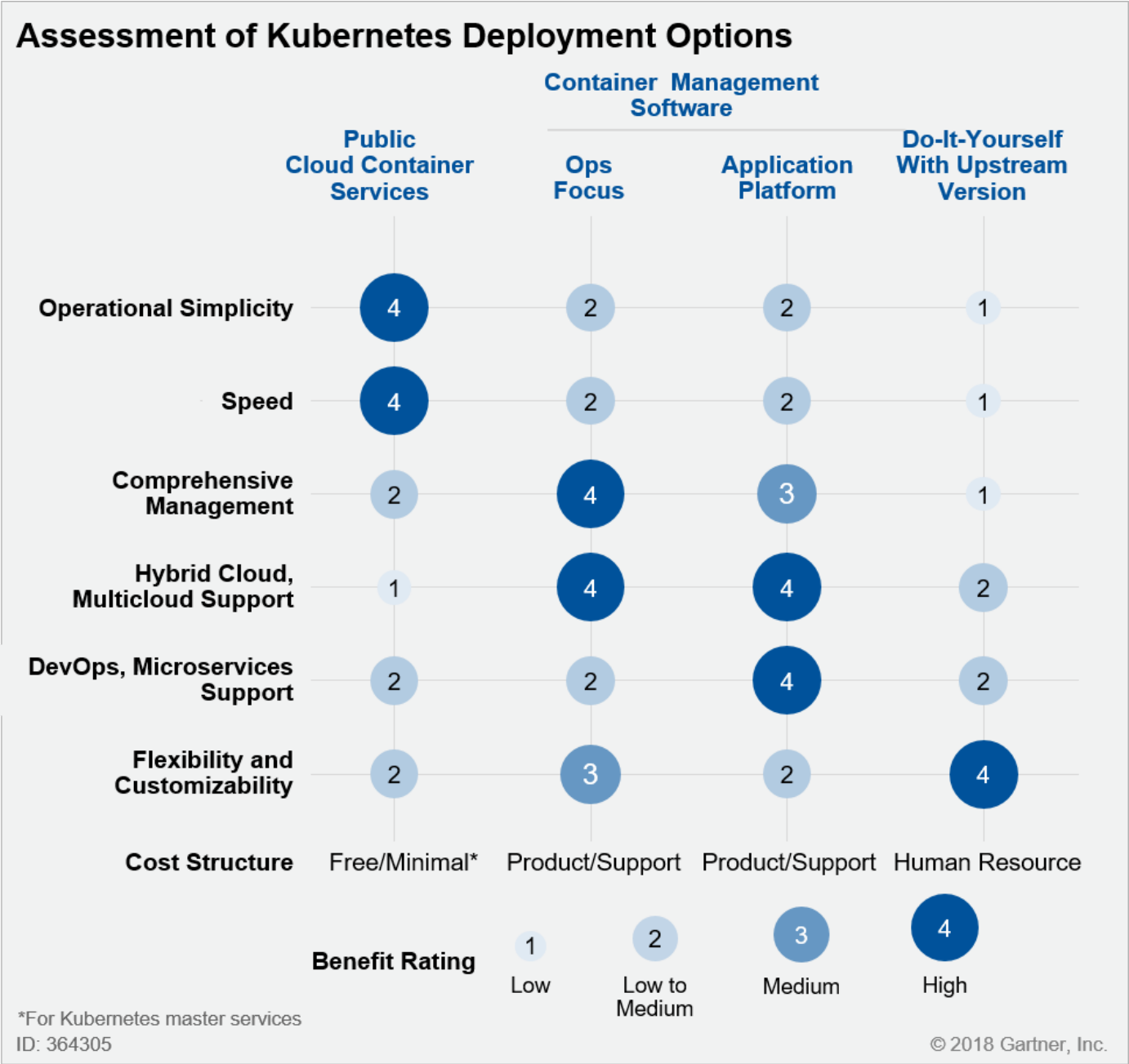
Recommendations:

- Collaborate with application leaders to identify your organization's priorities in deploying Kubernetes, based on the seven criteria.
- Use the priorities to evaluate deployment models explained in the next section.

Understand the Trade-Offs of Each Deployment Model

Each deployment model of Kubernetes from do-it-yourself to software solutions and cloud services has significant implications on costs, risks and skills required. Figure 2 shows Gartner's assessment of the deployment models, including a benefit rating on a scale of 1 to 4 based on the seven criteria.

Figure 2. Gartner's Assessment of Kubernetes Deployment Models



Source: Gartner (August 2018)

Each deployment model has pros and cons, so there's no “single right answer” to the question of how to adopt Kubernetes, but it is important to understand each deployment model’s trade-offs as shown in Table 2.

Table 2: Trade-Offs of Deployment Models

Deployment Model ↓	Benefits ↓	Challenges ↓
Public Cloud Container Services	Operational Simplicity: With this model, users don't have to build or manage Kubernetes clusters.*Cloud service providers are responsible for operations like	Hybrid Cloud, Multicloud Support: The public cloud container services focus on management of Kubernetes clusters that exists within each public cloud. If users want a Kubernetes cluster outside the public cloud, they have to deploy it separately. Though still in early phases,

Deployment Model ↓	Benefits ↓	Challenges ↓
	software upgrade, patching or failure recovery. Although this model does not eliminate the need for learning Kubernetes and containers, it will considerably minimize the learning curve. *In some cases, users have to manage Kubernetes worker node clusters. Speed: This model allows users to create or remove Kubernetes clusters on demand through public clouds' GUI, command line interface or API. Now that all major public cloud providers offer the managed services of Kubernetes, users can choose a public cloud provider commonly used by their organizations. Cost Structure: Although Kubernetes worker node clusters and serverless host management services (if you use it) come with a cost, using Kubernetes master services is free of charge or at a low price. (Google Kubernetes Engine [GKE] and Azure Kubernetes Service [AKS] are free; Amazon Elastic Container Service for Kubernetes [Amazon EKS] is 20 cents	several cloud providers have started offering on-premises software products that can potentially improve the hybrid cloud experience in the near future, (e.g., GKE On-Prem, IBM Cloud Private). Flexibility and Customizability: Although this point is being improved, many configurations in Kubernetes are tied to providers' native services or not controllable (e.g., Kubernetes version, networking, storage, logging). The troubleshooting method is also limited because you can't access low layers of infrastructure.

Deployment Model ↓	Benefits ↓	Challenges ↓
	per hour for each Kubernetes cluster.)	
Container Management Software	Comprehensive Management: Container management software simplifies managing containers in production systems by providing comprehensive management features not built into Kubernetes, such as security, governance, monitoring, logging and storage/networking integration. Hybrid Cloud, Multicloud Support: Container management software provides features to manage multiple Kubernetes clusters across a hybrid cloud or multicloud environment. <i>Container Ops Focus</i>-type products also occasionally federate public cloud container services, such as EKS/AKS/GKE, and on-premises clusters, which further simplifies building and managing Kubernetes clusters in a hybrid cloud or multicloud environment. <i>Application Platform</i>-type products also include service broker functionalities, based on Open Service Broker API.	Speed: Container management software simplifies building and managing Kubernetes clusters by automation and integration, but I&O still has to be familiar to a series of operations of Kubernetes cluster. That takes much more time than public cloud container services. <i>Container Ops Focus</i>-type products might be able to alleviate this issue by federating public cloud container services. Flexibility and Customizability: <i>Application Platform</i>-type products include vendor-specific or more opinionated components for better management or better experience of DevOps and microservices development. However, these narrow the flexibility and customizability to choose other tools. Cost Structure: Users need to pay product and support costs. Since public cloud container services are free (for Kubernetes master services) or more affordable, container management software usually costs considerably higher.

<https://www.gartner.com/doc/reprints?id=1-5EYO9EK&ct=180906&st=sb> 8/14

Deployment Model ↓	Benefits ↓	Challenges ↓
Do-It-Yourself With Upstream Version	Flexibility and Customizability: Users can choose operation tools and also customize Kubernetes code very easily. This allows both I&O and application developers to test new features in Kubernetes in a more timely way. This model provides the largest possibility to differentiate the container management platform from competitors by adding new features and capabilities through customization of Kubernetes and other related components.	Operational Simplicity: Typically, several dedicated engineers familiar with Kubernetes are necessary to design and maintain the DIY Kubernetes platform. Kubernetes still lacks common and agreed-upon tooling, plug-ins and patterns for an application platform. (See “Using Kubernetes to Orchestrate Container-Based Cloud and Microservices Applications.”) Engineers who can interact with the open-source community are often required for integration and troubleshooting (see “Achieving Kubernetes Operational Readiness to Run Containers in Production”). Speed: Much more careful planning, evaluation and testing are necessary compared with the other options. Users also need to build scripts and recipes for automation in deployment and updates. It usually takes more than half a year in production to accumulate knowledge as an organization. Cost Structure: Unless organizations already have experts who know Kubernetes well, costs are much higher than the other options because this model requires dedicated, more skillful human resources. As an exception, in very large deployments such as very large enterprises, this option may be able to deliver massive cost savings by customizing and optimizing the operation.

Source: Gartner (August 2018)

Other Deployment Options

Besides the above, there are deployment options worth considering:

- **Managed Services by Third-Party Providers:** Some third-party vendors provide SaaS-based operational management services for container deployments, independent of major public clouds (for example, Giant Swarm, Platform9). These managed services can support hybrid

cloud and multicloud environments better than the public clouds' services. Benefits of these services are similar to container management software, but provide more simplified operations and better speed than software with narrower flexibility and customizability.

- **Hosted Services:** Some vendors provide hosted services using container management software on public cloud environments like Amazon Web Services; for example, Red Hat OpenShift Dedicated and VMware Kubernetes Engine. Benefits of these services are similar to container management software, but provide more simplified operations by offloading operational tasks to the providers.

"Market Guide for Public Cloud Container Services" contains details of these two options.

Other Potentially Important Criteria

Consider these criteria as well:

- **Private cloud requirement:** Some organizations still forbid the use of public cloud for sensitive use cases, which requires a private cloud. If private cloud is a requirement, container management software, third-party providers of managed services supporting private cloud deployment, or hosted services are generally recommended.
- **Geolocation:** Public cloud container services generally make service implementation at various geographic locations easier. However, not every region of public clouds offer the container services, so deployment is still not flexible in many cases.

Recommendations:

- Understand the trade-offs of each deployment model for your organization by using Gartner's assessment result adjusted according to your organization's priorities and contexts.

Choose the Deployment Models According to Your Organization's Maturity

In this section, we share Gartner's views on which models are generally suitable for what kind of situations, along a scenario from evaluation to production phases.

Evaluation Phase

Public Cloud Container Services Are Generally Recommended as an Initial Choice

If your organization doesn't have staff with Kubernetes experience, public cloud container services are generally recommended as an initial choice at the evaluation phase, such as proofs of concepts:

- Avoid spending a lot of time learning Kubernetes operations and preparing infrastructure resources before using Kubernetes.
- No upfront investment. The cloud service model allows your organization to start testing Kubernetes with consumption-based pricing for nodes that host application containers.

Most public cloud container services are based on the “vanilla” Kubernetes open-source version, so you can keep interoperability at the Kubernetes level. That allows you to relatively easily extend or switch to other deployment models later, for example when you start operating Kubernetes in production.

Production Phase

Without Hybrid or Multicloud Requirements, Public Cloud Container Services Are Recommended

If your container platform can heavily depend on a specific public cloud (in other words, hybrid cloud or multicloud is not necessary), public cloud container service would be recommended in production as well.

Though management features are not comprehensive yet, every public cloud service provider is integrating their management and platform services one after another with their container service. If your container platform is only within, or almost within, a specific public cloud, this option will most likely make management easy. It is also a great advantage to be able to leverage the rapid evolution of hyperscale public cloud services and the benefits of a wide range of ecosystems.

Container Management Software Is the Best Choice in Production Phase If You Have Hybrid or Multicloud Requirements

When enterprises move from the early evaluation phase to the enterprisewide production phase, container management software is also often a reasonable choice because:

- You can simply integrate comprehensive management capabilities and better support hybrid cloud or multicloud use cases, which are often necessary to run production systems enterprisewide.
- Enterprise use of Kubernetes is still nascent, lacking best practices for repeatable, reliable operations. This model provides good guardrails.
- The application platform-type of container management software provides more comprehensive DevOps and microservices development experience.

Third-Party Managed Services and Hosted Services Can Alleviate Operational Burden

If the operational burden of container management software is a big challenge for your organization, managed services by third-party providers or hosted services are good alternatives to container management software. Third-party providers’ managed services enable more simplified operations and better speed than software, with narrower flexibility and customizability. Hosted services provide more simplified operations by offloading operational tasks to the providers.

Avoid Upstream Versions Unless There Are Special Requirements

To become operationally sound for production, Kubernetes requires coupling with many third-party plug-in components, creating significant infrastructure integration work (see “Assessing

Infrastructure Requirements for Deploying Kubernetes”). Integrating multiple open-source components with Kubernetes in a custom manner and maintaining your own system without vendor support is very tough. This model is only viable for organizations that can interact with and contribute to open-source projects.

However, even if your organization has human resources who can interact with Kubernetes and other open-source communities, Gartner recommends you avoid building your own Kubernetes system with upstream unless your organization has special customization requirements that market solutions cannot fulfill. As an exception, in very large deployments such as very large enterprises, this option may be able to achieve significant cost savings by customizing and optimizing the operation.

Recommendations:

- Leverage the cloud’s advantages of operational simplicity, speed and cost by considering using public cloud container services as an initial choice. If hybrid cloud or multicloud is a requirement at the production phase, consider extending or switching to other deployment models.
- Consider using container management software at the enterprisewide production phase to better support hybrid cloud and multicloud use cases, to efficiently integrate management capabilities not built in Kubernetes and to better support DevOps and microservices development.
- Consider managed services by third-party providers or hosted services as good alternatives to container management software if the operational burden of container management software is a big challenge for your organization.
- Avoid building your own Kubernetes system with upstream unless there are special requirements that market solutions cannot fulfill.

Acronym Key and Glossary Terms

Amazon EKS	Amazon Elastic Container Service for Kubernetes
AKS	Azure Kubernetes Service
CNCF	Cloud Native Computing Foundation (https://www.cncf.io/)
DevOps	DevOps (https://www.gartner.com/it-glossary/devops) represents a change in IT culture, focusing on rapid IT service delivery through the adoption of agile, lean practices in the context of a system-oriented approach.
GKE	Google Kubernetes Engine

Note 1

Kubernetes Vendor Ecosystem

This list is not meant to illustrate all vendors involved in the Kubernetes ecosystem. It is meant to show they are active in this area:

- Alibaba
- Amazon Web Services
- Cisco
- Diamanti
- DH2i
- Docker
- Giant Swarm
- Google
- Hashicorp
- Heptio
- IBM
- Joyent
- Kublr
- Mesosphere
- Microsoft
- Mirantis
- Oracle
- Pivotal
- Platform9
- Rancher Labs
- Red Hat
- SUSE

■ VMware

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